

Muhammad Faheem

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EXPERIENCE

National Centre for Physics (NCP) Research Intern

Islamabad, Pakistan
July 2024 – July 2025

- Conducted a comparative analysis of classical and deep learning-based object tracking methodologies, with primary focus on single-object tracking (SOT) architectures, frameworks, and evaluation protocols in research-driven environments
- Researched deep learning-based fault detection and anomaly detection techniques for HVAC systems, leading to an accepted IEEE conference paper titled *“Deep Learning-Based Fault Detection Framework for Predictive Maintenance in HVAC Systems”* at the 13th IEEE Conference on Technologies for Sustainability (SusTech 2026)

National Aerospace Science & Technology Park (NASTP) AI Engineering Intern

Rawalpindi, Pakistan
June 2025 – August 2025

- Designed and implemented an LLM-assisted OCR pipeline for business card digitization, enabling structured entity extraction (names, roles, organizations, contact details) and integration with an internal contact management system
- Conducted a research-focused implementation of an offline, gRPC-based multi-person behavior analysis system, integrating object detection, multi-object tracking (MOT), human pose estimation, and spatio-temporal action recognition for controlled evaluation and benchmarking

PROJECTS

Driver Assistance System Using RT-DETR

- Designed a real-time ADAS pipeline using fine-tuned RT-DETR, achieving **+18.6% mAP@50** over baseline DETR on dashcam-style data, with end-to-end inference latency of **3845 ms/frame** on RTX-class GPUs
- Deployed a production-grade inference stack using FastAPI (ASGI) and Next.js, supporting **1080p video streams at 25 FPS** with REST-based analytics endpoints

Multi-person Behaviour Recognition System

- Implemented an offline, modular gRPC pipeline integrating YOLO-based detection, DeepSORT-style MOT, pose estimation, and spatio-temporal action recognition for **simultaneous tracking of 10+ subjects per frame**
- Optimized inter-service communication and batching, reducing pipeline latency by **31%** and enabling **sub-120 ms end-to-end processing** on multi-camera video sequences

Action Segmentation Using MS-TCN

- Implemented an MS-TCN architecture in PyTorch for frame-level video action segmentation, improving segmental F1@50 by **+14.2%** compared to single-stage TCN baselines
- Trained and evaluated on long untrimmed videos (**10k+ frames/sequence**), achieving stable convergence and **sub-5 ms/frame inference** during evaluation

EDUCATION

National University of Computer and Emerging Sciences B.S. Artificial Intelligence

Islamabad, Pakistan
Aug 2022 – Present

- Honors: Deans List (Spring 2023, Fall 2025) | GPA: 3.21/4.00
- Relevant Coursework: Generative AI, Agentic AI, Reinforcement Learning, Computer Vision, ANN, Machine Learning, Digital Image Processing, NLP, Data Structures, Algorithms

SKILLS

Programming & Frameworks: Python, C/C++, JavaScript, TypeScript, FastAPI, React/Next.js

AI & ML: PyTorch, TensorFlow, Keras, scikit-learn, YOLO, ONNX, TensorRT, Hugging Face, LangChain, LangGraph, n8n

Data & Systems: NumPy, Pandas, Git, Docker, Linux, AWS, GCP, MCP